

Physical Security Evaluation of ASCON-128: Deep-Learning Side-Channel Analysis and Clock-Glitch Fault Injection

Prajwal R • Prajwal G S Basavaraj • Sachin S • Nitesh
Dept. of Electronics and Communication Engineering, PES University, Bengaluru, India
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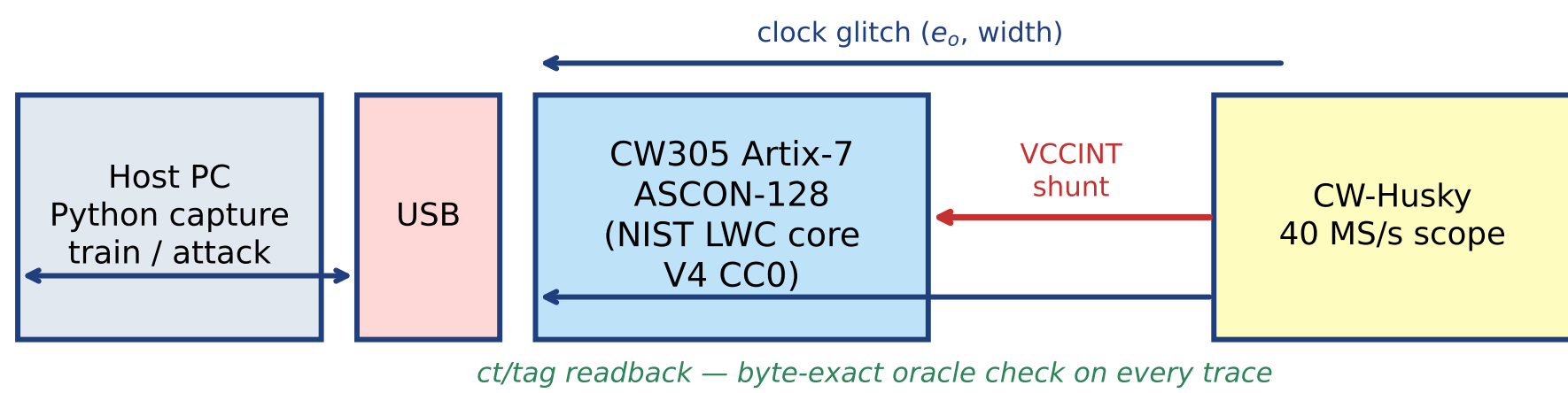
Contributions

- First hardware-measured DL-SCA + clock-glitch FI on the official unmasked NIST LWC ASCON-128 core (ascon-verilog V4) on CW305/CW-Husky — prior FPGA work attacked custom cores or software implementations only.
- Per-column key-bit recovery demonstrated on live hardware: on-board fine-tuning lifts a profile from the majority floor to 90.7% (72% floor); the fine-tuned column’s closed-loop attack recovers its 2 key bits in 32 queries at posterior 0.999.
- Five leakage-validation gates separating genuine side-channel signal from five classes of confident false positives — two artifact classes unreported for LWC hardware.

Why ASCON-128 on Real Hardware?

ASCON-128 is the NIST SP 800-232 lightweight cryptography standard, entering constrained FPGA deployments. Its security under physical adversaries — power analysis and fault injection — determines whether the standard’s promise holds in the field. Published FPGA attacks evaluate custom ASCON cores (SCARL) or software implementations (ACNS’24 ensemble) — never the official unmasked LWC core. We measure that core directly.

Oracle-Verified Testbed



- CW305 Artix-7 + CW-Husky, 40 MS/s vs 10 MHz crypto clock (4 samples/cycle), VCCINT shunt, per-query trigger.
- Byte-exact oracle gate: every stored trace accepted only when its ct/tag readback matches a software reference (5/5 NIST KATs). Kills misconfigured-target errors before they poison datasets.
- Masked (d=1, 2-share) reference core on the same platform — the unmasked-vs-masked contrast isolates masking as the causal variable.

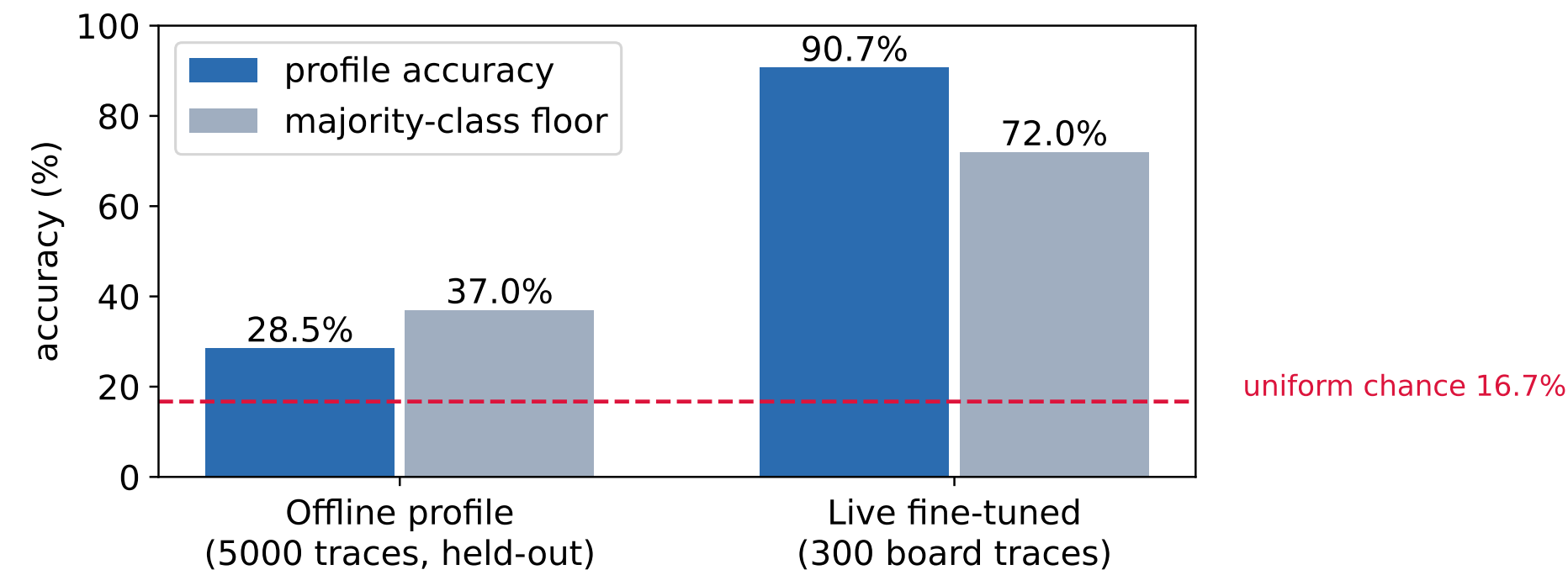
DL-SCA Setup: Divide-and-Conquer

One CNN profile per S-box column: each of the 64 columns carries 2 unknown key bits $\Rightarrow$ 4 hypotheses per column. Train on random-key traces (80/20 split), HW labels 0–5. Random keys are essential: at a fixed key the column label is a function of two public nonce bits (see Gates, trap 3).

Signal Preprocessing Pipeline

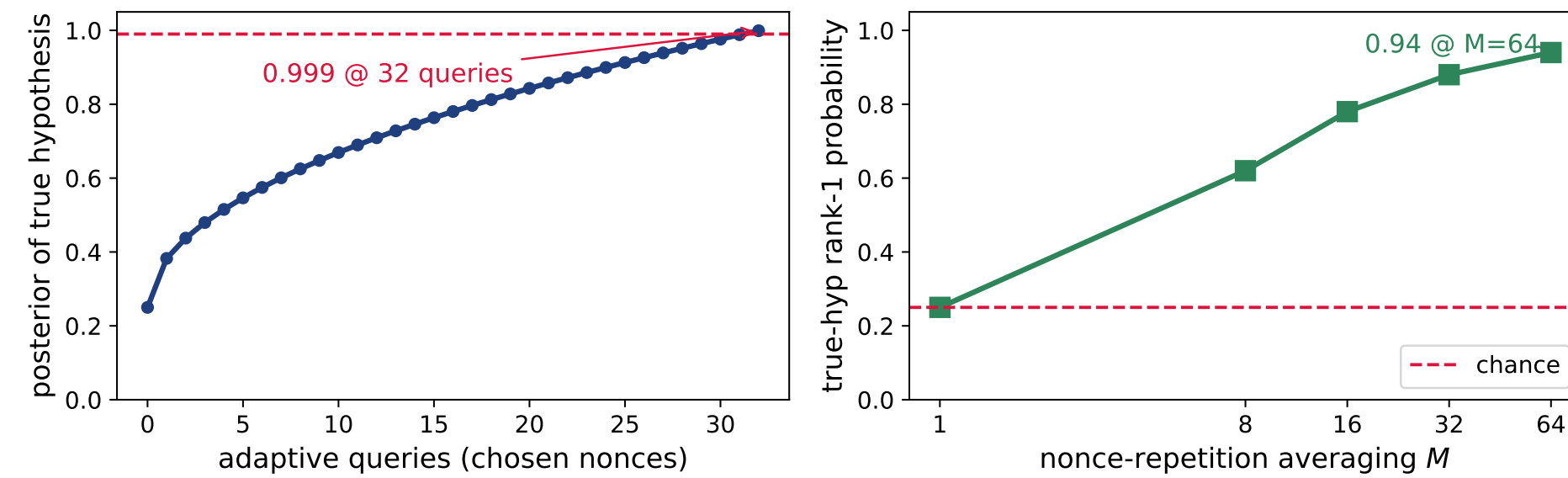
- Sample-exact alignment: phase-locked cross-correlation against an unmasked reference trace removes scope sampling jitter.
- Trace-level standardization: per-sample z -score normalization $(t - \mu)/\sigma$ cancels DC drift and board temperature variance.
- Targeted temporal cropping: active round-1 window isolation focuses the convolutional receptive field strictly on physical leakage.

Result 1: Domain Gap & Live Fine-Tuning



- The profiling-domain gap: offline profiles trained on raw traces face analog SNR and capture-transfer drops, remaining near the majority floor ($\approx 37\%$).
- Live on-board fine-tuning closes the gap: 300 board traces at a known key (random nonces) lift classifier accuracy to 90.7% vs. 72.0% majority floor (+18.7% margin).
- First-order signal recovered: the fine-tuned profile reliably identifies true key bits on live hardware where raw models fail.

Result 2: Closed-Loop Recovery + SNR Lever



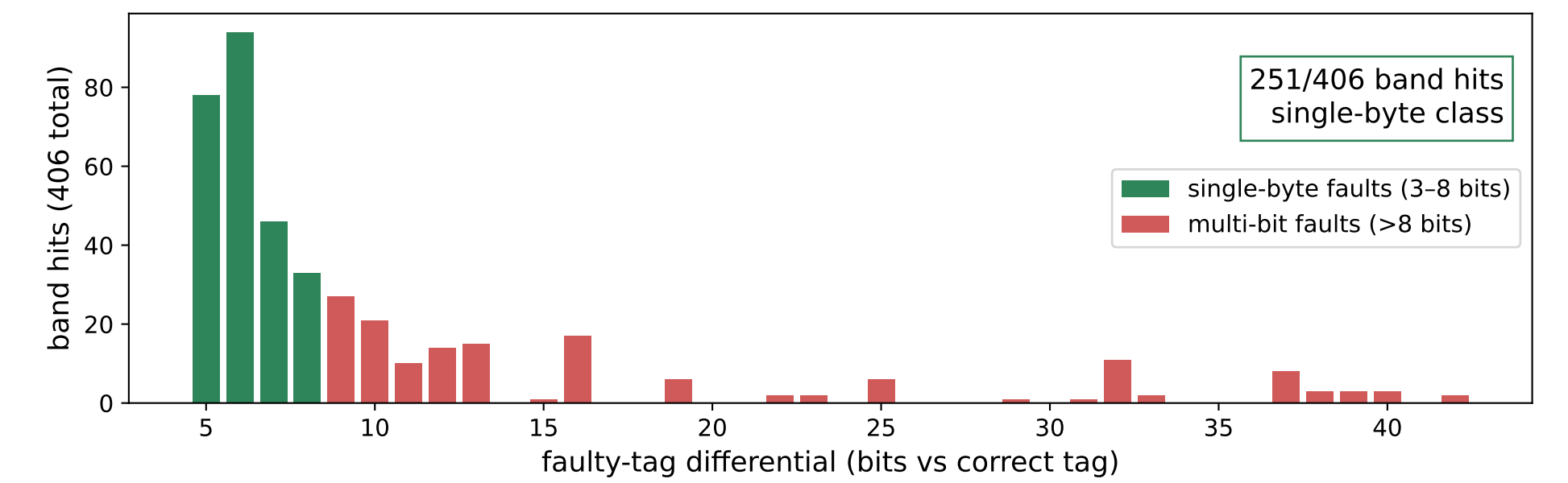
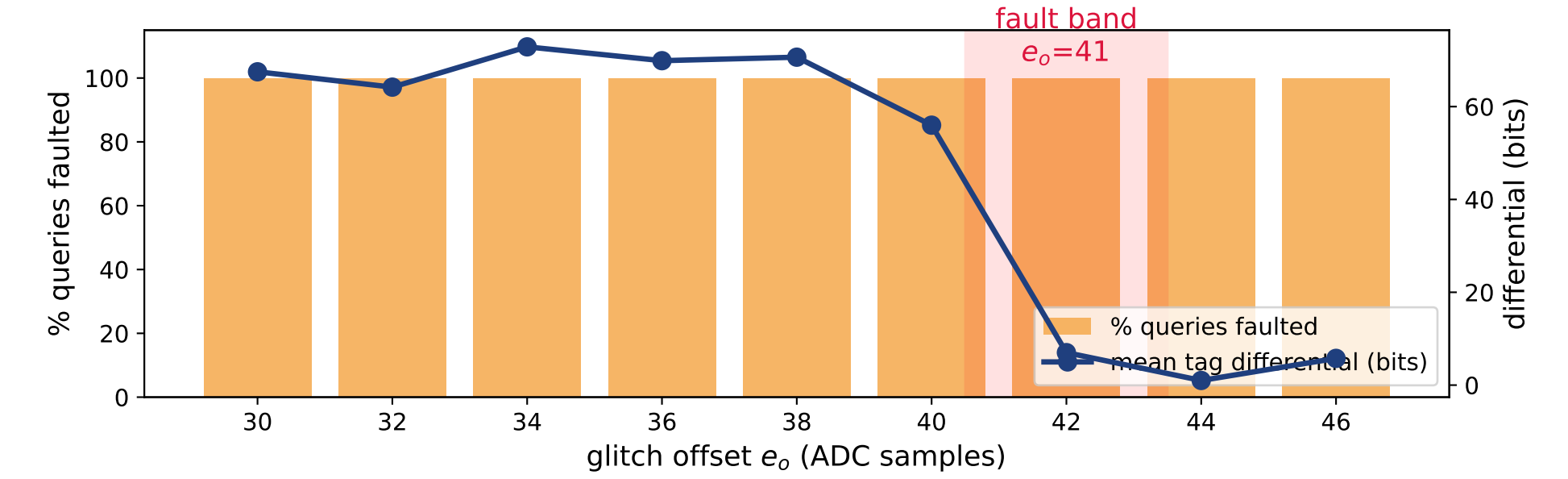
- Adaptive closed-loop recovery: posterior-scoring chosen-nonce loop converges to the correct 2 key bits in 32 queries (posterior 0.999) using public nonces and power traces only.
- Trace averaging ($\sqrt{M}$ SNR lever): repeating a separating nonce M times lifts true-hypothesis rank-1 from 0.25 (chance) to 0.94 at $M=64$, and accelerates convergence from 15 to 9 queries per column.
- Full 128-bit key assembly: divide-and-conquer per column + bounded 2^m enumeration over weak columns + re-encryption verification. Pipeline verified; scaling live fine-tuning to all 64 columns is ongoing work.

Full-Key Assembly Strategy

- Divide-and-conquer recombination: independently evaluate the posterior distribution $\Pr(h \mid \mathbf{T}_c)$ for all 64 columns.
- Confidence-guided candidate ranking: sort columns by log-posterior margin $\Delta \log p = \log p_{(1)} - \log p_{(2)}$; prioritize high-certainty columns while maintaining top-2 hypotheses for marginal columns.
- Bounded verification: brute-force search over k ambiguous columns (2^{2k} space) with immediate physical re-encryption gating against fresh nonces.

Take-home: The unmasked NIST core leaks its key bits column-by-column; $\sqrt{M}$ averaging and on-board fine-tuning are the levers that turn weak per-trace signal into confident recovery.

Clock-Glitch Fault Injection



- A stable fault band at $e_o=41$ (width 67): 406 band hits over 153 distinct tags; differentials 27–41 bits.
- 12 clean single-bit round-11 faults localized to exact (column, bit) — 10 distinct state columns — each verified by exact faulty-tag match under simulated single-bit flip.
- Data-selected placement: each nonce faults exactly one column, reproducibly $\Rightarrow$ DFA coverage bounded at one column/query. Nonce-set engineering (≈ 266 nonces, coupon-collector bound) restores coverage.
- Burst glitches: 400/400 faulted but multi-bit (30+ bits) — not composable.

Masked Core (d=1, 2-share)

Same pipeline, 9,151 traces: S-box target at chance for every architecture. First-order masking works; cost $\approx 1.8\times$ throughput, $2.1\times$ LUTs.

Five Leakage-Validation Gates

Artifact trap	Failure mode	Gate
1. Prior-driven “edge”	edge measures the class prior, not leakage	held-out acc > majority floor
2. Prior-only profile + separating-nonce picker	0.993 posterior on the wrong hypothesis	profiles must clear the floor
3. Public-nonce shortcut	model reads nonce-load activity, not key bits	train on random keys only
4. Key-reload bus artifact	$ r \approx 0.6$ fake leakage from key-writes; 64/64 fake columns	survive fixed-key, varying nonces
5. Structured-key alignment	ramp key fakes 75.8% recovery; inverted key scores higher	inversion-invariance + null controls

Traps 4–5 are, to our knowledge, unreported for LWC hardware evaluation.

References

- [1] NIST, SP 800-232: ASCON Lightweight Cryptography Standards, 2023.
- [2] S. Picot et al., “SCARL,” J. Cryptographic Engineering, 2021.
- [3] Rezaeezade et al., “One for All, All for Ascon,” ACNS, 2024 (ePrint 2023/1922).